Molecular structure inference from molecular spectra

with transformers, eventually

Multimodal NMR $\rightarrow$ Molecular structure

based on the ~800'000 molecule synthetic dataset

[1] Marvin Alberts et al. "Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry". In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url]

in turn based on

[2] Daniel Lowe. *Chemical reactions from US patents (1976–Sep2016)*. figshare, 2017. [url]

and MestReNova software (NMR prediction plug-in)

GPT-2 reference implementation[1]

Input: $x \in \mathbb{N}^{B \times T}$ (token indices)

Token embedding: $\mathbf{E}_{\text{tok}} = \text{Embedding}(x) \in \mathbb{R}^{B \times T \times d}$

Position embedding: $\mathbf{E}_{\text{pos}} = \text{Embedding}[1 : T]$

Combined: $\mathbf{H}^{(0)} = \text{Dropout}(\mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}})$

For each block $\ell = 1 \ldots L$:

$\mathbf{X} = \mathbf{H}^{(\ell-1)}$

$\mathbf{X} \leftarrow \mathbf{X} + \text{CausalSelfAttn}(\text{LayerNorm}(\mathbf{X}))$

$\mathbf{X} \leftarrow \mathbf{X} + \text{MLP}(\text{LayerNorm}(\mathbf{X}))$

$\mathbf{H}^{(\ell)} = \mathbf{X}$

Final: $\mathbf{H}_{\text{final}} = \text{LayerNorm}(\mathbf{H}^{(L)})$

Logits: $\hat{y} = \mathbf{H}_{\text{final}} \cdot \mathbf{E}_{\text{tok}}^{\top}$ (weight tying)

[1] Based on https://github.com/karpathy/nanoGPT/blob/master/model.py

Roadmap

Phase I – Pretraining

- ✓ Pretrain on SMILES inputs, and:
  - ✓ bias a hidden state towards ChemBERTa-v2,
  - ✓ hijack a hidden state to predict functional groups,
  - ✓ augment with non-canonical SMILES.
- Correctness self-assessment flag – after SMILES add ✓/✗
- Backtracking / inference-time reasoning – provide/teach *undo*
- Pretrain on the one-atom-one-peak "quasi-NMR"

Phase II – NMR to SMILES inference training

- ✓ More careful data split (correlations due to USPTO origin)
- Compare vs Alberts et al., 2025 [3]
- Compare H+C+HSQC vs H+HSQC

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Phase II – NMR to SMILES inference training

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Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][␣][∘][␣][⇄]C([O-])([O-])=O.[Ca+2]

A given SMILES structure

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Unsupervised trigger token (molecular embedding)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][⁝][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Supervised hidden state (molecular embedding)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

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O=C([O-])[O-].[Ca+2][⁚][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Supervised hidden state (functional groups)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Unsupervised trigger token (alt SMILES)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][⁝][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Distribution-supervised alternative SMILES

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][∘][_][⇄]C([O-])([O-])=O.[Ca+2]

Supervised “end of sequence”

Phase I – Pretraining: losses definitions

- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log\det T + \frac{n}{2}\log 2\pi)$ – embedding ($n = 383$)
- $\frac{1}{n}(\|T\mathbf{e}\|_1 - \log\det T + n\log 2)$ – functional groups ($n = 289$)
- KL divergence, position-averaged – next-token distribution
- Causal LM $= -\log p(\text{eos} \mid \text{prefix})$ – end-of-sequence token

combined à la Kendall et al. [4]:

$$\text{minimize} \quad \sum_t \left(\frac{1}{\sigma_t^2}\text{loss}_t + \log(\sigma_t^2)\right)$$

with a trainable σ_t for each task t.

Phase I – Pretraining: training setup

Run $1/3$:

- 580'738 training samples, 72'189 validation samples (top PubChem)
- Max. SMILES length: 50 characters; "multimodal dataset" excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 64
- Optimizer `adabelief` with learning rate 10^{-4}
- No embedding or functional groups supervision
- One of five alt-SMILES per sample per epoch

[code] [weights]

Phase I – Pretraining: training setup

Run 2/3 (resumed from Run 1):

- 688'876 training samples, 85'873 validation samples (top PubChem)
- Max. SMILES length: 70 characters; "multimodal dataset" excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 64
- Optimizer `adabelief`, learning rate 10^{-4}
- With embedding and functional groups supervision
- One of five alt-SMILES per sample per epoch

[code] [weights]

Phase I – Pretraining: training setup

Run 3/3 (resumed from Run 2):

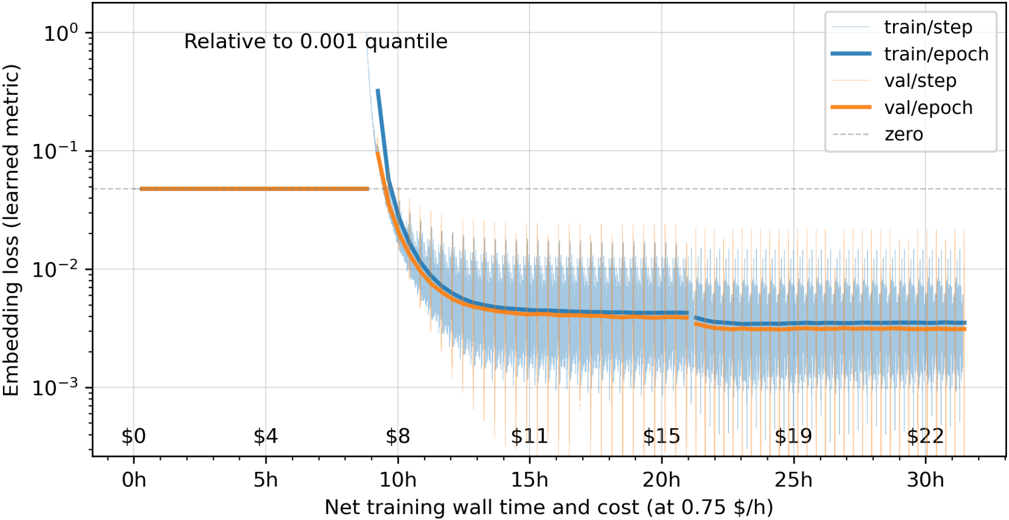
- 688'876 training samples, 85'873 validation samples (top PubChem)
- Max. SMILES length: 70 characters; "multimodal dataset" excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 96
- Optimizer `adabelief`, learning rate 10^{-4}
- With embedding and functional groups supervision
- One of five alt-SMILES per sample per epoch

[code] [weights]

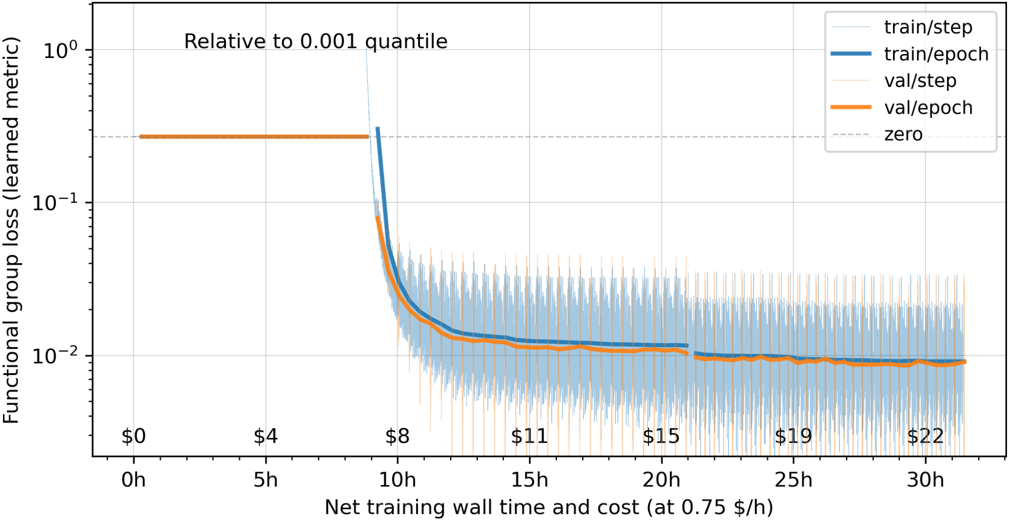
ONE ETERNITY LATER...

Losses

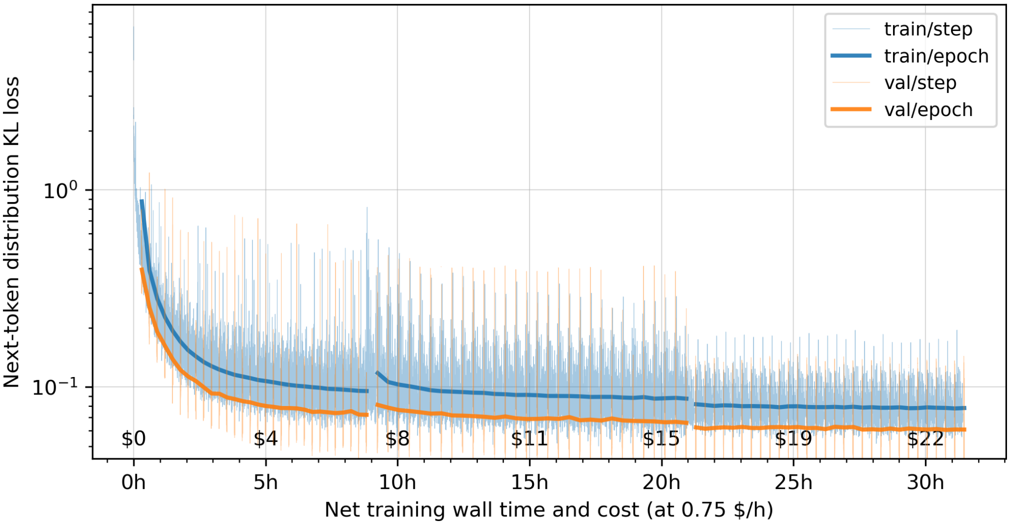
during training



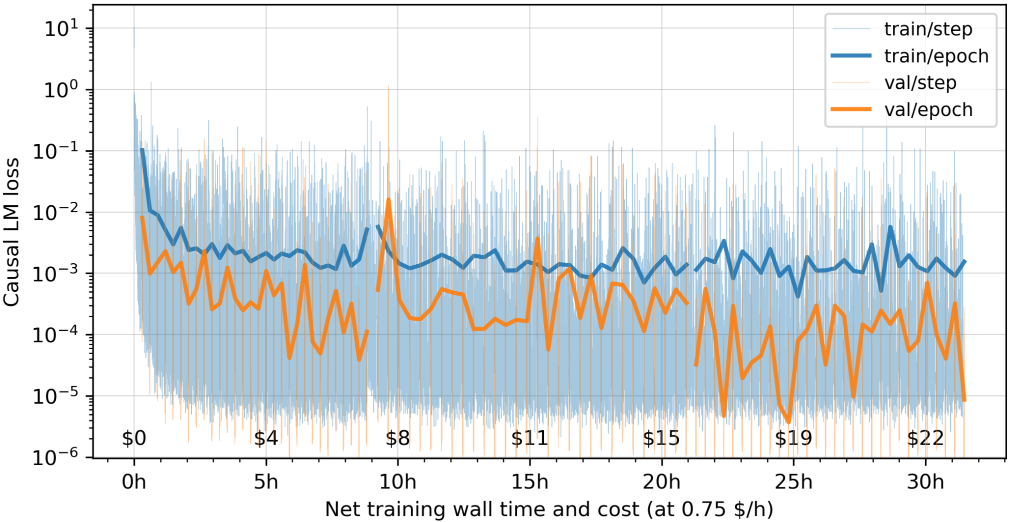
Relative to 0.001 quantile
train/step
train/epoch
val/step
val/epoch
zero
10^0
10^{-1}
10^{-2}
10^{-3}
Embedding loss (learned metric)
$0
$4
$8
$11
$15
$19
$22
0h
5h
10h
15h
20h
25h
30h
Net training wall time and cost (at 0.75 $/h)



Relative to 0.001 quantile
train/step
train/epoch
val/step
val/epoch
zero
Functional group loss (learned metric)
10^0
10^{-1}
10^{-2}
$0
$4
$8
$11
$15
$19
$22
0h
5h
10h
15h
20h
25h
30h
Net training wall time and cost (at 0.75 $/h)



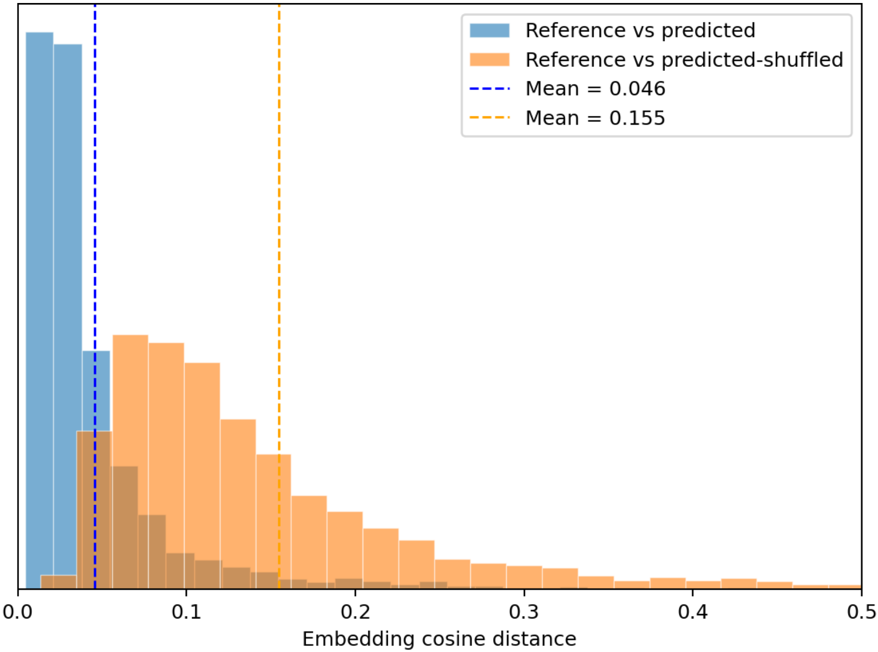
train/step
train/epoch
val/step
val/epoch
Next-token distribution KL loss
10^0
10^{-1}
$0
$4
$8
$11
$15
$19
$22
0h
5h
10h
15h
20h
25h
30h
Net training wall time and cost (at 0.75 $/h)



train/step
train/epoch
val/step
val/epoch
Causal LM loss
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
$0
$4
$8
$11
$15
$19
$22
0h
5h
10h
15h
20h
25h
30h
Net training wall time and cost (at 0.75 $/h)

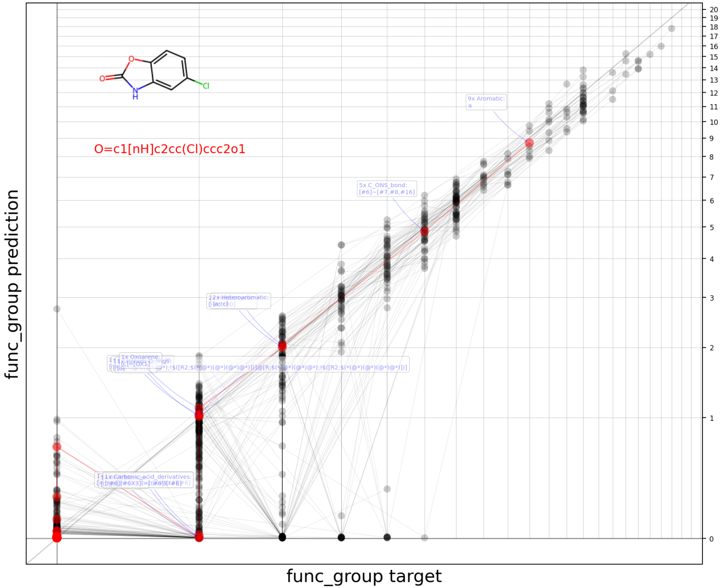
ChemBERTa-2 embedding

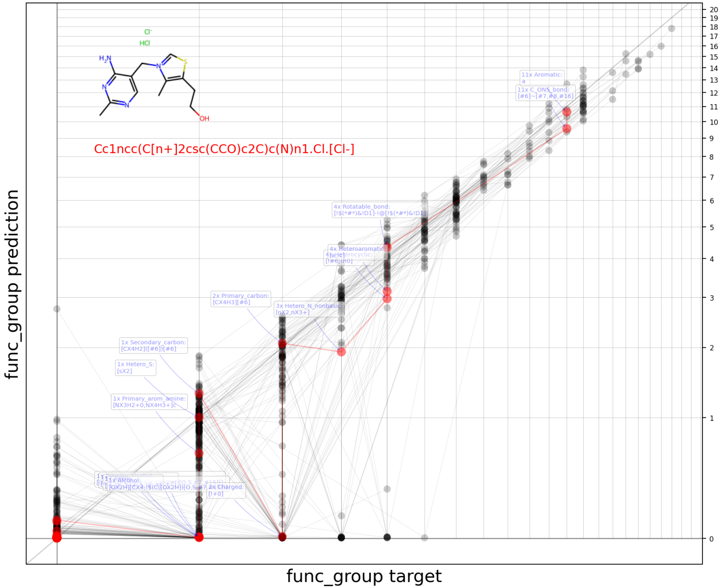
384-d vector

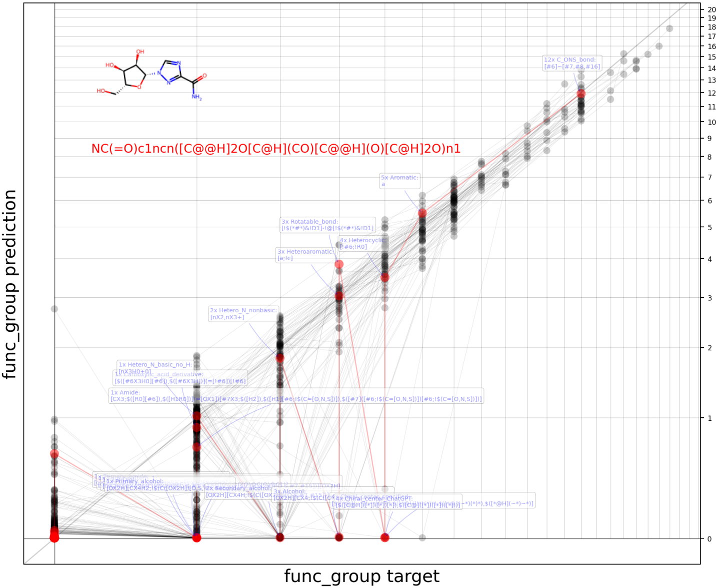

Reference vs predicted
Reference vs predicted-shuffled
Mean = 0.046
Mean = 0.155
0.0
0.1
0.2
0.3
0.4
0.5
Embedding cosine distance

Functional groups

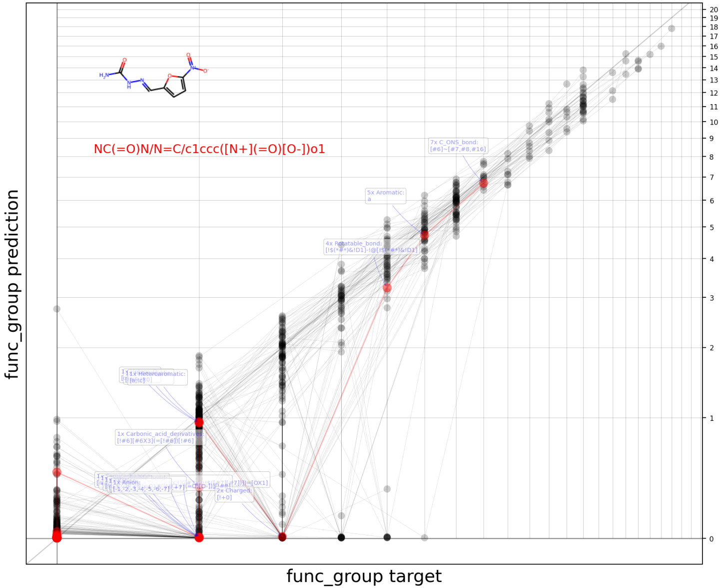
289-d vector of counts


O=c1[nH]c2cc(Cl)ccc2o1
9x Aromatic:
a
5x C_ONS_bond:
[#6]~[#7,#8,#16]
2x Heteroaromatic:
1x Oxoarene:
[c]=[OX1]
1x Carbonic_acid_derivatives:
func_group prediction
func_group target


Cc1ncc(C[n+]2csc(CCO)c2C)c(N)n1.Cl.[Cl-]
11x Aromatic:
a
11x C_ONS_bond:
[#6]~[#7,#8,#16]
4x Rotatable_bond:
[!$(*#*)&!D1]-!@[!$(*#*)&!D1]
4x Heteroaromatic:
[a;!c]
2x Primary_carbon:
[CX4H3][#6]
3x Hetero_N_nonbasic:
[nX2,nX3+]
1x Secondary_carbon:
[CX4H2]([#6])[#6]
1x Hetero_S:
[sX2]
1x Primary_arom_amine:
[NX3H2+0,NX4H3+]c
1x Alcohol:
2x Charged:
[!+0]
func_group prediction
func_group target



NC(=O)c1ncn([C@@H]2O[C@H](CO)[C@@H](O)[C@H]2O)n1
12x C_ONS_bond:
[#6]~[#7,#8,#16]
5x Aromatic:
a
3x Rotatable_bond:
[!$(*#*)&!D1]-!@[!$(*#*)&!D1]
4x Heterocyclic:
[!#6;!R0]
3x Heteroaromatic:
[a;!c]
2x Hetero_N_nonbasic:
[nX2,nX3+]
1x Hetero_N_basic_no_H:
[nX3H0+0]
1x Amide:
1x Primary_alcohol:
2x Secondary_alcohol:
3x Alcohol:
4x Chiral_center_ChatGPT:
func_group prediction
func_group target



NC(=O)N/N=C/c1ccc([N+](=O)[O-])o1
func_group prediction
func_group target
7x C_ONS_bond:
[#6]~[#7,#8,#16]
5x Aromatic:
a
4x Rotatable_bond:
1x Heteroaromatic:
1x Carbonic_acid_derivatives:
[!#6][#6X3](=[!#6])[!#6]
1x Anion:
2x Charged:
[!+0]

SMILES augmentation

supervision of next-token distribution

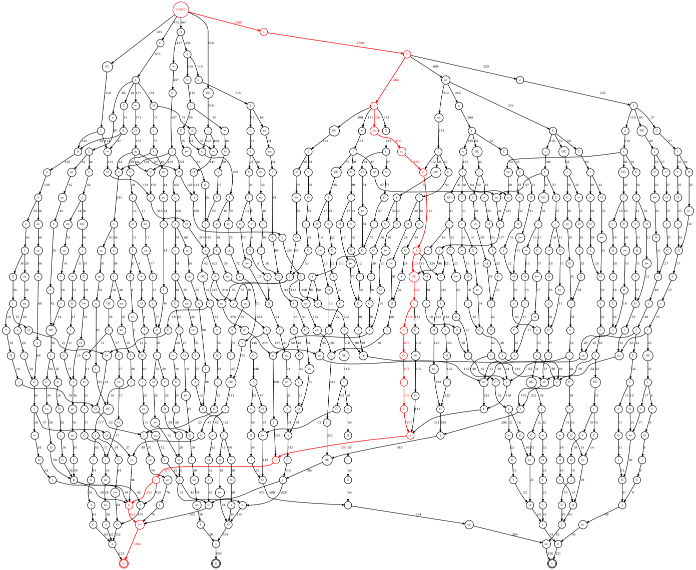
SMILES augmentation

Aspirin can be represented as:

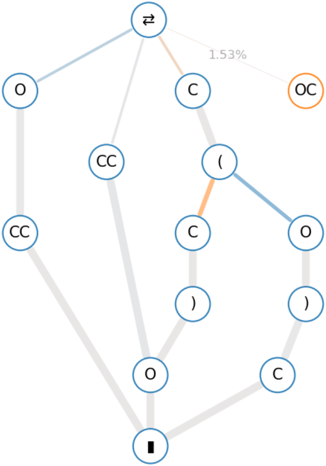
- O(c1ccccc1C(O)=O)C(=O)C
- c1(c(OC(C)=O)cccc1)C(=O)O
- c1cc(OC(C)=O)c(C(=O)O)cc1
- ...

304 variants total (unkekulized)

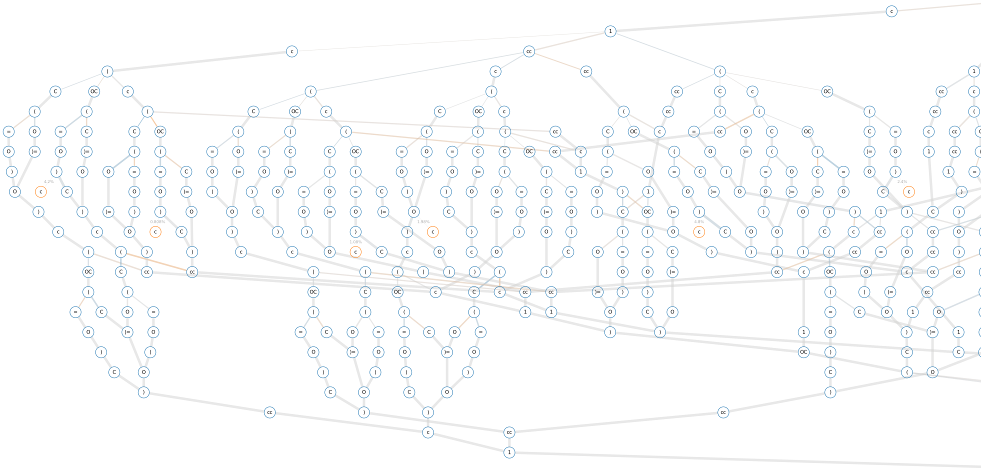
SMILES augmentation



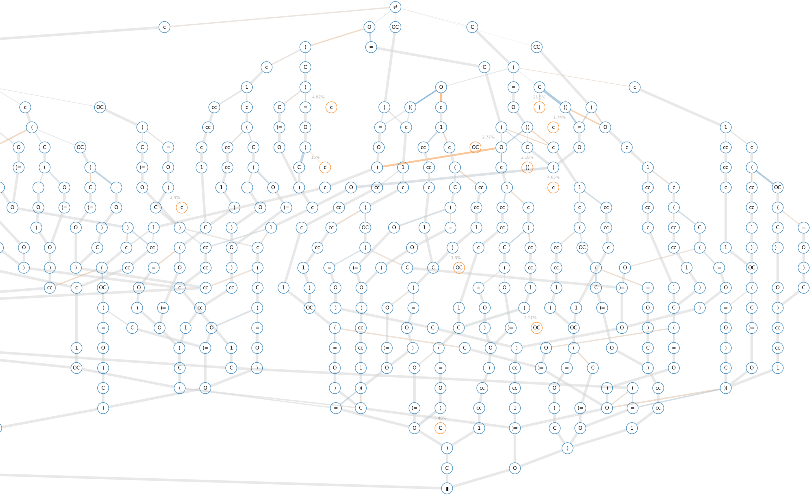
SMILES augmentation: ethanol



SMILES augmentation: aspirin (out-of-sample)



SMILES augmentation: aspirin (out-of-sample)



Are SMILES correct?

out-of-sample

Input SMILES: `CC(C)(C)c1ccc(C(=O)NCc2ccco2)cc1`

predicted variant	score	valid	correct	isomer
`c1cc(C(C)(C)C)ccc1C(=O)NCc1occc1`	-0.18	✓	✓	✓
`c1c(ccc(c1)C(C)(C)C)C(=O)NCc1occc1`	-0.18	✓	✓	✓
`c1c(ccc(C(C)(C)C)c1)C(=O)NCc1occc1`	-0.18	✓	✓	✓
`c1cc(ccc1C(C)(C)C)C(=O)NCc1occc1`	-0.18	✓	✓	✓
`c1c(C(C)(C)C)ccc(c1)C(=O)NCc1occc1`	-0.18	✓	✓	✓
`c1cc(C(C)(C)C)ccc1C(NCc1occc1)=O`	-0.19	✓	✓	✓
`c1cc(C(C)(C)C)ccc1C(=O)NCc1ccco1`	-0.19	✓	✓	✓
`CC(c1ccc(cc1)C(=O)NCc1occc1)(C)C`	-0.19	✓	✓	✓
`c1c(ccc(C(C)(C)C)c1)C(=O)NCc1ccco1`	-0.19	✓	✓	✓
`c1c(ccc(c1)C(C)(C)C)C(=O)NCc1ccco1`	-0.19	✓	✓	✓

Input SMILES: `CC(C)C1CCC(Cc2ccc(Cl)cc2)C1(O)Cn1cncn1`

predicted variant	score	valid	correct	isomer
`c1cc(ccc1Cl)CC1CCC(C1(O)Cn1cncn1)C(C)C`	-0.18	✓	✓	✓
`c1cc(ccc1CC1CCC(C(C)C)C1(Cn1cncn1)O)Cl`	-0.19	✓	✓	✓
`c1cc(Cl)ccc1CC1CCC(C(C)C)C1(Cn1cncn1)O`	-0.19	✓	✓	✓
`c1cc(ccc1Cl)CC1CCC(C(C)C)C1(Cn1cncn1)O`	-0.2	✓	✓	✓
`c1cc(ccc1Cl)CC1CCC(C(C)C)C1(O)Cn1cncn1`	-0.2	✓	✓	✓
`c1cc(Cl)ccc1CC1CCC(C(C)C)C1(O)Cn1cncn1`	-0.2	✓	✓	✓
`CC(C)C1CCC(Cc2ccc(cc2)Cl)C1(O)Cn1cncn1`	-0.2	✓	✓	✓
`CC(C)C1CCC(Cc2ccc(cc2)Cl)C1(Cn1cncn1)O`	-0.2	✓	✓	✓
`c1c(ccc(c1)Cl)CC1CCC(C(C)C)C1(Cn1cncn1)O`	-0.2	✓	✓	✓
`c1cc(ccc1Cl)CC1CCC(C(C)C)C1(O)Cn1ncnc1`	-0.22	✓	✓	✓

Input SMILES: `COC(=O)C1=CO[C@@H](C)[C@@H]2CN3CCc4c([nH]c5cc(OC)ccc45)[C@@H]3C[C@H]12`

predicted variant	score	valid	correct	isomer
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@@H]21)C`	-0.24	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@H]21)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@@H]21)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@@H]12)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@H]12)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@H]21)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@@H]12)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@H]12)C`	-0.25	✓	✗	✓
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]2C1)C`	-0.26	✗	✗	✗
`O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]2C1)C`	-0.26	✗	✗	✗

Input SMILES: N#N

predicted variant	score	valid	correct	isomer
N#N	-0.038	✓	✓	✓
C(#N)N	-0.3	✓	✗	✗
c1(ccccn1)N	-0.74	✓	✗	✗
C(#N)CN	-0.87	✓	✗	✗
c1c(N)#N	-0.95	✗	✗	✗
c1(ccccn1)#N	-0.97	✗	✗	✗
c1(cccc)N	-1	✗	✗	✗
C(N)#N	-1.1	✓	✗	✗
c1(N)cc	-1.1	✗	✗	✗
C(#N)CC	-1.1	✓	✗	✗

Paper summary: Alberts et al., 2025 [3]

[3] Marvin Alberts et al. *Automated structure elucidation at human-level accuracy via a multimodal multitask language model*. 2025. [url]

"Top–1 accuracies of up to 96%" on the "Chemical Education dataset"

[5] Scott E. Van Bramer and Loyd D. Bastin. "Spectroscopy data for undergraduate teaching". In: *Journal of Chemical Education* 100.10 (2023), pp. 3897–3902. [url]

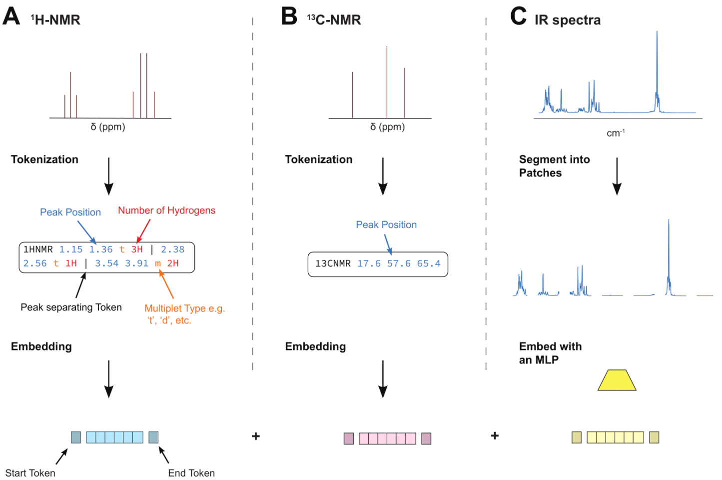


Figure 2: Combining different spectroscopic modalities. A and B) Tokenisation procedure for ^{1}H-NMR and ^{13}C-NMR spectra respectively. C) IR spectra are segmented into patches, and each patch is projected into the embedding space via an MLP. Bottom: The embeddings of each modality are flanked with a modality start and modality end token and concatenated. The concatenation of all modalities is fed into the model.

Step 1 – Synthetic pre-training

- 6-layer encoder–decoder transformer
- pretrained on 790'000 paired IR + C/H NMR spectra from [1]

Step 2 – Multitask architecture

- A single model handles every modality and their combinations
- Each modality block is wrapped in *modality-start* / *modality-end* tokens

Step 3 – Baseline finetune (171 paired spectra)

- Five-fold CV on the Chemical Education dataset
- Multimodal top-1: 69.10 ± 11.16; top-5: 91.47 ± 5.58

Modality	Multi	Simulated		Zero-shot		Finetuned	
		Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
IR	✗	39.72	57.28	1.05	4.21	17.97±5.65	44.37±6.62
C	✗	49.35	66.85	48.42	66.32	57.75±11.26	85.97±9.62
H	✗	60.08	76.74	20.00	31.58	45.83±8.14	72.89±4.21
IR+C+H	✗	73.40	87.83	17.89	33.68	50.04±11.25	81.26±9.08
IR	✓	33.57	53.21	2.11	11.58	15.97±8.65	47.14±8.30
C	✓	41.37	61.80	53.16	82.11	58.57±9.71	89.61±4.69
H	✓	53.44	72.82	31.58	51.58	49.96±5.99	79.22±3.91
IR+C+H	✓	73.05	88.69	50.53	61.05	69.10±11.16	91.47±5.58

Table: Table 1 from Alberts et al., 2025 [3]

Step 4 – +600 unpaired spectra

- Added 600 molecules with only IR and/or 13C data
- IR: NIST EPA gas–phase library; C-NMR: NMR-ShiftDB2
- Multimodal Top-1 rises to 75.41 ± 8.32

Step 4' – +33'088 unpaired spectra

- Further augmentation: 29'960 $\times$ C, 5'027 $\times$ IR, 1'899 dual-modality
- IR: NIST EPA gas–phase library; C-NMR: NMR-ShiftDB2
- Multimodal Top-1 rises to 96.25 ± 7.50

Modality	**Finetuned**		**+600 Samples**		**+33,088 Samples**	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
IR	15.97±8.65	47.14±8.30	21.69±6.41	52.86±9.33	42.50±12.12	75.00±6.85
C	58.57±9.71	89.61±4.69	63.55±5.84	90.55±5.37	96.25±7.50	97.50±5.00
H	49.96±5.99	79.22±3.91	52.77±3.66	78.23±7.27	93.75±12.50	96.75±2.50
IR+C+H	69.10±11.16	91.47±5.58	75.41±8.32	91.52±1.87	96.25±7.50	98.75±2.50

Table: Table 2 from Alberts et al., 2025 [3] thx.

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